

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1142

COURSE OUTCOME COVERED

CO4:

implement object-oriented system layers using design principles and patterns, and integrate access and view layers with OODBMS (BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 1

Total Marks:10

Annexure A

Industry Problem Solving Task

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Industry Problem Solving Task

Question 1: E-Commerce Shopping Platform

Patterns: MVC Pattern | Singleton Pattern | DAO Pattern

A mid-sized e-commerce company is building a scalable shopping platform to handle thousands of users daily. The system follows a layered object-oriented architecture, where the presentation layer provides web and mobile interfaces, the business logic layer processes orders, payments, and inventory, and the access layer communicates with an Object-Oriented Database Management System (OODBMS) to store product and customer data.

To enhance flexibility, the system uses MVC pattern for UI separation, Singleton pattern for managing database connections, and DAO pattern for data access. The design ensures that new features like discount modules can be added without modifying existing code. Performance is improved using caching and efficient query handling.

Question: Analyze how design principles, layered architecture, and design patterns are applied in this system to ensure scalability and maintainability, and explain how the layers are integrated with OODBMS.

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

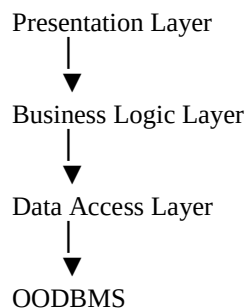
Question 1: E-Commerce Shopping Platform

Introduction

The E-Commerce Shopping Platform is designed using **Object-Oriented Design Principles**, **Layered Architecture**, and **Design Patterns** to support thousands of users efficiently. The system ensures scalability, maintainability, flexibility, and efficient database management through integration with an **Object-Oriented Database Management System (OODBMS)**.

1. Layered Architecture

The system follows a **three-layer architecture**:



A. Presentation Layer

Provides user interaction through:

- Web Interface
- Mobile Application

Responsibilities

- Display products
- Manage shopping cart
- Accept user requests
- Show order status and payment details

Components

- User Interface (UI)
- Controllers
- Views

B. Business Logic Layer

Handles all business operations.

Responsibilities

- Order Processing
- Payment Processing
- Inventory Management
- Discount Calculation
- Customer Validation

Example Classes

- OrderService
- PaymentService
- InventoryService
- DiscountService

Benefits

- Centralized business rules
- Easy maintenance
- Better scalability

C. Data Access Layer

Acts as an interface between business logic and database.

Responsibilities

- Store customer data
- Retrieve product information
- Update inventory
- Save orders

Example DAO Classes

- ProductDAO
- CustomerDAO
- OrderDAO
- InventoryDAO

2. Application of Object-Oriented Design Principles

Encapsulation

Data and methods are grouped together within classes.

Example

```
Product
-----
productID
productName
price

getPrice()
updatePrice()
```

Benefits

- Data security
- Controlled access
- Easy maintenance

Inheritance

Common features are inherited from a base class.

```
User
^
-----
|         |
| Customer | Admin
```

Benefits

- Code reuse
- Reduced duplication

Polymorphism

Different classes implement the same method differently.

```
Payment
^
-----
|         |
| CardPayment | UPIPayment
```

Method:

processPayment()

Benefits

- Extensibility
- Flexible implementation

Open/Closed Principle (OCP)

The system is open for extension but closed for modification.

Example

Discount

^

| |
FestivalDiscount PremiumDiscount

New discount types can be added without changing existing classes.

Benefits

- Easy feature addition
- Reduced risk of bugs

3. MVC Pattern

MVC separates application components into Model, View, and Controller.

Model

Represents business data and logic.

Examples:

- Product
- Customer
- Order

View

Represents user interface.

Examples:

- Product Page
- Shopping Cart Page
- Payment Page

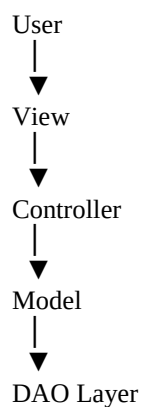
Controller

Handles user requests and connects Model with View.

Examples:

- ProductController
- OrderController
- PaymentController

MVC Workflow



Advantages

- Separation of concerns
- Easy maintenance
- Reusable UI components

4. Singleton Pattern

The Singleton Pattern is used for database connection management.

Structure

```
DatabaseConnection
|
+---- getInstance()
```

Only one database connection object exists throughout the application.

Advantages

- Reduced memory usage
- Better performance
- Consistent database access
- Avoids multiple unnecessary connections

5. DAO (Data Access Object) Pattern

DAO separates database code from business logic.

DAO Classes

```
ProductDAO
CustomerDAO
OrderDAO
InventoryDAO
```

Functions

```
addProduct()
getProduct()
updateProduct()
deleteProduct()
```

Advantages

- Loose coupling
- Database independence
- Easier maintenance
- Better code organization

6. Integration with OODBMS

The Object-Oriented Database stores objects directly rather than converting them into relational tables.

Integration Flow

Presentation Layer



Business Logic Layer



DAO Layer



Singleton Database Connection



OODBMS

Objects Stored in OODBMS

- Product Objects
- Customer Objects
- Order Objects
- Inventory Objects
- Payment Objects

Benefits of OODBMS

- Direct object storage and retrieval
- Faster access to complex data
- Eliminates object-relational mapping issues
- Better support for object-oriented applications

7. Scalability and Performance Features

Caching

Stores frequently accessed data.

Examples:

- Product catalog
- Popular products
- Customer sessions

Benefits

- Faster response time
- Reduced database load

Efficient Query Handling

- Indexed searches
- Optimized DAO methods
- Reduced redundant queries

Benefits

- Improved performance
- Better scalability for thousands of users

Advantages of the Overall Design

Feature	Benefit
Layered Architecture	Separation of responsibilities
MVC Pattern	Clear UI separation
Singleton Pattern	Efficient resource management
DAO Pattern	Structured database access
OODBMS	Direct object storage
OCP Compliance	Easy feature extension
Caching	Improved performance
Query Optimization	Better scalability

Conclusion

The E-Commerce Shopping Platform achieves **scalability, maintainability, and flexibility** through the effective use of **Object-Oriented Design Principles, Layered Architecture, and Design Patterns**. The **MVC Pattern** separates user interface components, the **Singleton Pattern** efficiently manages database connections, and the **DAO Pattern** provides clean access to the OODBMS. Integration with the OODBMS enables direct storage and retrieval of objects, while caching and optimized queries enhance performance. Together, these design approaches create a robust, scalable, and maintainable e-commerce solution capable of handling thousands of users daily.